

Ex-no: 7	SHARED MEMORY IPC
17/03/2026	

Aim

To implement **Inter Process Communication (IPC)** using **Shared Memory** in C, where one process writes data into shared memory and another process reads the same data.

Algorithm

Writer Process Algorithm

1. Start the program.
2. Generate a unique shared memory key using `ftok()`.
3. Create a shared memory segment using `shmget()`.
4. Attach the shared memory segment using `shmat()`.
5. Write data/message into shared memory.
6. Display the written message.
7. Detach the shared memory using `shmdt()`.
8. Stop the program.

Reader Process Algorithm

1. Start the program.
2. Generate the same shared memory key using `ftok()`.
3. Access the existing shared memory segment using `shmget()`.
4. Attach the shared memory using `shmat()`.
5. Read the message from shared memory.
6. Display the received message.
7. Detach the shared memory using `shmdt()`.
8. Remove the shared memory segment using `shmctl()`.
9. Stop the program.

CODE:

Writer.c

```
#include<stdio.h>
```

```
#include<sys/ipc.h>

#include<sys/shm.h>

#include<string.h>


int main()

{

    key_t key;

    int shmid;

    char *str;


    // Generate unique key

    key = ftok("shmfile",65);


    // Create shared memory

    shmid = shmget(key,1024,0666|IPC_CREAT);


    // Attach shared memory

    str = (char*) shmat(shmid,NULL,0);


    // Write data

    printf("Write Data: ");

    fgets(str, 1024, stdin);


    printf("Data written in memory: %s\n",str);
```

```
// Detach shared memory  
  
shmdt(str);  
  
return 0;  
}
```

Reader.c

```
#include<stdio.h>  
  
#include<sys/ipc.h>  
  
#include<sys/shm.h>  
  
  
int main()  
{  
  
    key_t key;  
  
    int shmid;  
  
    char *str;  
  
  
    // Generate same key  
  
    key = ftok("shmfile",65);  
  
  
    // Access shared memory  
  
    shmid = shmget(key,1024,0666);  
  
  
    // Attach shared memory  
  
    str = (char*) shmat(shmid,NULL,0);
```

```
// Read data

printf("Data read from memory: %s\n",str);


// Detach shared memory

shmdt(str);


// Destroy shared memory

shmctl(shmid,IPC_RMID,NULL);


return 0;
}
```

OUTPUT:

Write Data: Hello Shared Memory IPC

Data written in memory: Hello Shared Memory IPC

Data read from memory: Hello Shared Memory IPC

Result

Thus, the program for **Inter Process Communication using Shared Memory** was implemented successfully, and data was transferred between processes using shared memory.